Wildfire Effects on AQI: What We Learned and How to Use It

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Case Study 1 Recap

- Analyzed traffic, meteorology, and temporal patterns.
- Key finding: AQI variation largely explained by traffic and weather.
- Linear models performed best for forecasting.

Case Study 1 Limitations

- The Baseline Gap: Case 1 models performed well on average but failed to capture episodic “extreme AQI” spikes.
- The Risk: The days where our baselines fail are the exact days our users rely on us most for health advisories.
- The Hypothesis: These aligned error spikes are driven by unobserved exogenous factors—specifically, wildfire smoke transport—rather than a lack of model complexity.

Business Question

The key business question is: How can we make next-day AQI forecasts more reliable during high-risk pollution events, particularly during wildfire smoke episodes?

More specifically:

Can wildfire activity explain extreme AQI spikes that current models fail to capture?

Can wildfire signals be integrated into the forecasting pipeline to improve advisory reliability?

Data

Data Source: Canadian Wildland Fire Information System (CWFIS) Fire M3 Daily Hotspots archive for 2022–2024.

- Filtered strictly to Ontario borders and stripped out records with missing coordinates or timestamps to guarantee accurate downstream spatial joins.
- Identified and removed physically impossible sensor errors, such as surface temperatures recorded above 100°C. Deliberately retained records where the Head Fire Intensity (HFI) was exactly 0.

Model Assumptions

We fit a Bayesian hierarchical regression model with a Gaussian likelihood to estimate the effect of wildfire exposure on next-day AQI.

Predictors:

- Wildfire exposure lags (0–3 days)
- Meteorology (temperature, precipitation, wind)
- Traffic intensity
- Weekday effects (day-of-week indicators)

Model Assumptions

Hierarchical structure

- Station-level random intercepts

- Cyclic seasonal spline for day-of-year

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	11.70	3.34	5.25	18.18	1.00	1683	1734
wsei_hfi_k0_z	1.20	0.23	0.74	1.64	1.00	3717	2883
wsei_hfi_k1_z	-0.67	0.31	-1.27	-0.04	1.00	3071	2613
wsei_hfi_k2_z	0.30	0.35	-0.36	1.00	1.00	2817	2446
wsei_hfi_k3_z	4.12	0.28	3.58	4.65	1.00	3495	3039
MeanTemp_z	0.61	0.82	-1.01	2.24	1.00	3812	3219
TotalPrecip_z	-1.77	0.15	-2.08	-1.47	1.00	4736	2878
MaxTemp_z	0.98	0.83	-0.68	2.58	1.00	3138	2787
MinTemp_z	0.27	0.79	-1.28	1.81	1.00	2724	2650
TempRange_z	1.94	0.32	1.32	2.57	1.00	2595	2564
TotalTraffic_z	-1.27	0.35	-1.97	-0.58	1.00	4579	2784
wind_dir_deg_z	-1.22	0.15	-1.51	-0.94	1.00	4856	2877

Bayesian Estimation

Fitting

- Posterior inference via MCMC (NUTS sampler)
- 4 chains × 2,000 iterations (per chain)

Posterior evidence

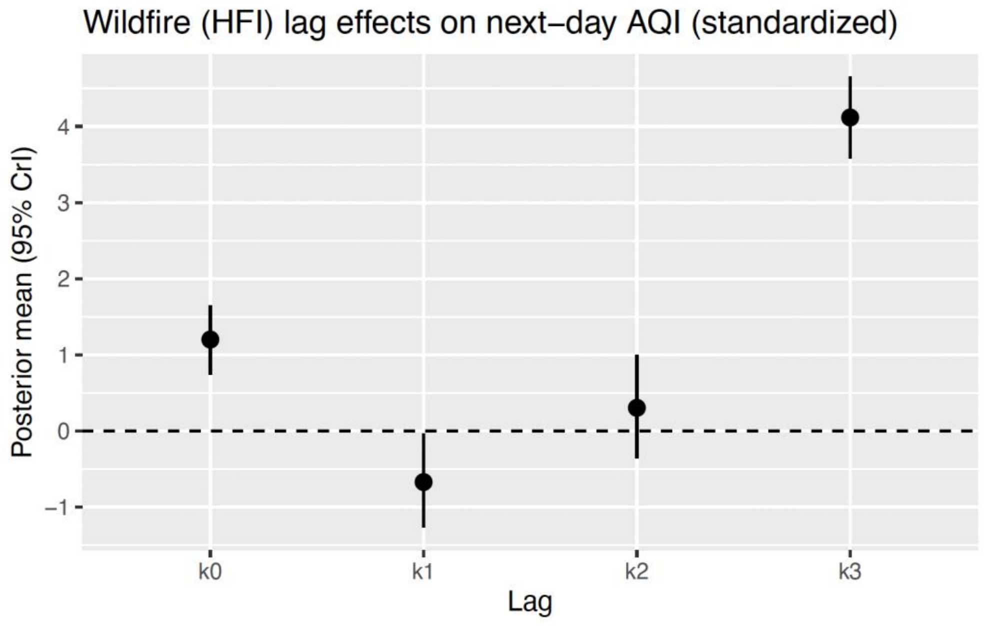
- The posterior probability that the cumulative 0–3 day wildfire effect is positive is > 99.9%

MCMC diagnostics

- R-hat ≈ 1.00 across parameters (good mixing / convergence)
- High effective sample sizes (ESS), suggesting estimates are stable and not driven by sampling noise

Wildfire Lag Effect

- Key takeaway: wildfire exposure is associated with higher next-day AQI, with effects concentrated in short lags (0–3 days).

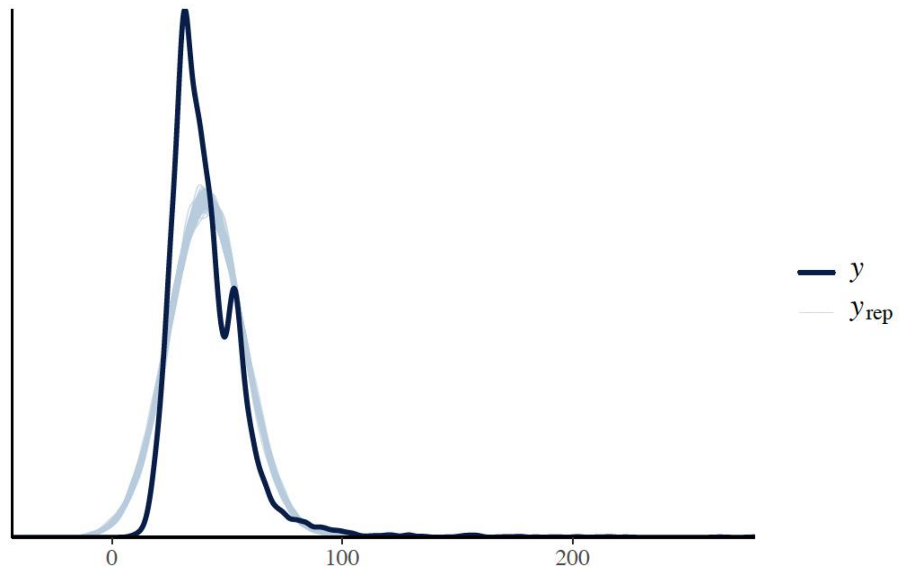


Cumulative Wildfire Effect

- Cumulative effect over lags 0–3 days:
  - Mean ≈ 4.95 AQI increase
  - 95% credible interval ≈ [4.63, 5.27]
  - Probability positive ≈ 1.

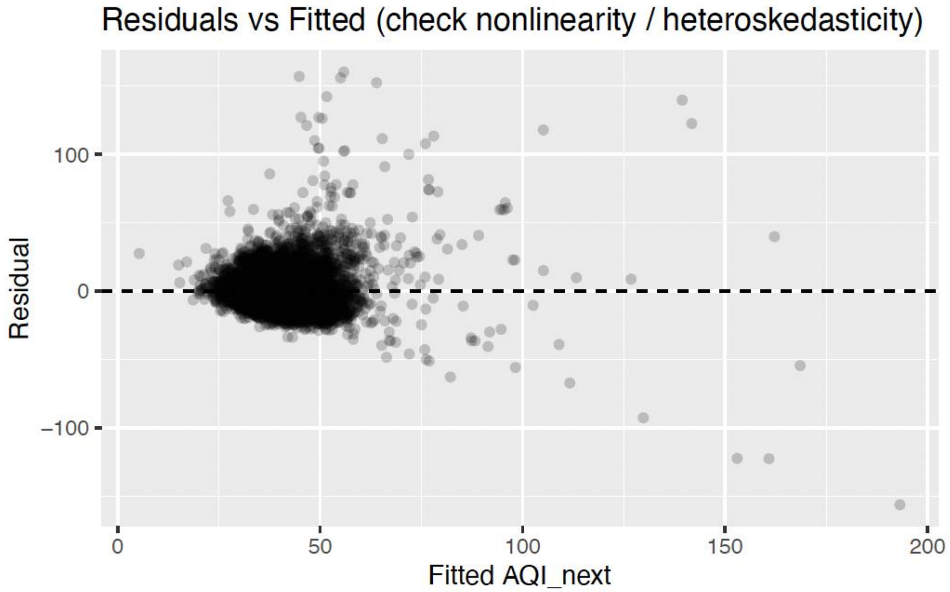
Posterior Predictive Check

- Model reproduces the main AQI distribution.
- Some mismatch in extreme right tail.
- Indicates difficulty modeling extreme events.



Residual Diagnostics

- Residual variance increases at higher AQI levels.
- Suggests heteroskedasticity during severe pollution days.



Heavy-Tail Modeling

- Student-t likelihood used to capture extreme pollution spikes.
- Estimated degrees of freedom ≈ 3.9.
- Indicates heavy-tailed AQI distribution.

Key takeaway:

- Strongest wildfire association occurs at lag 3, with a clearly positive 95% credible interval.

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	37.77	0.97	35.65	39.48	1.00	903	1596
wsei_hfi_k0_z	0.41	0.33	-0.21	1.08	1.00	5332	4688
wsei_hfi_k1_z	1.39	0.47	0.48	2.33	1.00	4393	4433
wsei_hfi_k2_z	-1.49	0.89	-3.20	0.14	1.00	3594	4321
wsei_hfi_k3_z	5.25	0.50	4.26	6.22	1.00	5884	4719
MeanTemp_z	0.64	0.82	-0.98	2.26	1.00	5597	4048
TotalPrecip_z	-1.66	0.12	-1.91	-1.42	1.00	8382	4630
MaxTemp_z	1.03	0.82	-0.57	2.64	1.00	4882	4461
MinTemp_z	0.37	0.77	-1.17	1.88	1.00	4035	4210
TempRange_z	1.84	0.30	1.25	2.43	1.00	4206	4131
TotalTraffic_z	-0.66	0.31	-1.27	-0.08	1.00	4906	4315
wind_dir_deg_z	-1.10	0.12	-1.34	-0.86	1.00	7462	4027

Gaussian vs Student-t

- Gaussian: thin tails → extremes inflate σ and can distort uncertainty.
- Student-t: heavy tails (small ν) → handles spikes without over-inflating σ.
- Robustness: wildfire lag effects (esp. lag 3) stay positive in both models; Student-t is often cleaner/stronger because outliers have less leverage.

Remaining Limitations

- wildfire hotspots are only proxies for smoke transport
- wind direction treated linearly
- lag coefficients may trade off due to correlation
- extreme AQI days remain hard to predict

Research Question 2

- RQ2 (Forecast lift): Does adding wildfire exposure features improve next-day AQI point-forecast accuracy,
- with emphasis on extreme days (e.g., top decile AQI) where Case 1 failed?

RQ2 Goal & Setup

- Question: Does adding wildfire exposure features improve next-day AQI point forecasts?
- Emphasis: extreme days (top-decile next-day AQI) where Case Study 1 failed
- We compare Baseline vs +Wildfire across three model families:
  - OLS
  - Neural Network (MLP)
  - LSTM

Handling Missing Values

Data sources include station outages and missing predictors.

Strategy:

- Drop rows with missing target AQI(t+1)
- For missing predictors:
    - simple imputation (e.g., forward fill / median by station) OR
    - complete-case filtering for model-ready rows

Report final sample size:

- Remaining observations: N = 18523 (original: 33496)
- Remaining stations: S = 21

Avoiding Data Leakage

- We fit the imputer only on the training data and then transform both training and testing data.

Result

overall performance comparison

	model_family	spec	RMSE	MAE
0	LSTM	+ Wildfire (cfg_2)	11.675581	8.716072
1	OLS	+ Wildfire	11.704901	8.697979
2	OLS	Baseline	11.726270	8.682664
3	LSTM	Baseline (cfg_2)	11.786356	8.939309
4	NN	Baseline	12.300621	9.295328
5	NN	+ Wildfire	12.782537	9.542219

Result

extreme days performance comparison

	model_family	spec	top_decile_RMSE	top_decile_MAE
0	NN	Baseline	21.523922	18.380774
1	OLS	Baseline	22.470814	19.058420
2	NN	+ Wildfire	22.558778	19.144850
3	OLS	+ Wildfire	22.638192	19.316394
4	LSTM	Baseline (cfg_2)	23.453293	20.651726
5	LSTM	+ Wildfire (cfg_2)	23.531045	20.583334

Interpretation

- Overall: OLS ≈ LSTM; NN worse
- Extreme days: NN baseline best; +Wildfire slightly worse (all models)
- Business: no strong smoke-day lift; avoid over-claim
- Action: deploy OLS baseline; wildfire for monitoring/explanation
- Next: transport-aware features; extreme-focused training

Research Question 3

- RQ3 (Operational signal):

Can we produce a stable, interpretable wildfire-risk signal (feature + diagnostics) that supports stakeholder-facing explanations and monitoring?

Define the Operational Signal

- Operational target: “High AQI tomorrow” event
- Definition: next-day AQI in the top 10% (binary outcome)
- Why this helps stakeholders: a single number for alerts + dashboards

Procedures

Step 1: Define the Monitored Outcome

Create a binary “high-AQI tomorrow” event (e.g., top 10% of next-day AQI).

Step 2: Fit an Interpretable Risk Model

Train a Bayesian hierarchical logistic regression.

The model’s wildfire exposure features, especially the 3-day lag, meaningfully increase the probability of a high-AQI day tomorrow.

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-2.55	0.15	-2.86	-2.25	1.00	879	1600
wsei_hfi_k0_z	-0.07	0.04	-0.16	0.01	1.00	3291	2997
wsei_hfi_k1_z	0.17	0.06	0.05	0.29	1.00	2653	2630
wsei_hfi_k2_z	-0.17	0.07	-0.31	-0.03	1.00	2522	2574
wsei_hfi_k3_z	0.45	0.07	0.32	0.59	1.00	3234	2895
MeanTemp_z	0.19	0.82	-1.45	1.75	1.00	3065	2535
TotalPrecip_z	-0.43	0.06	-0.55	-0.33	1.00	4545	3039
MaxTemp_z	0.34	0.84	-1.36	1.95	1.00	2661	2558
MinTemp_z	0.18	0.77	-1.34	1.68	1.00	2204	2462
TempRange_z	0.41	0.28	-0.15	0.97	1.00	2173	2105
TotalTraffic_z	-0.14	0.07	-0.29	0.01	1.00	2337	2352
wind_dir_deg_z	-0.26	0.04	-0.33	-0.19	1.00	4419	2527

Result: Wildfire Increases High-AQI Risk

- Focus on the most actionable term: lag-3 wildfire exposure (k3)
- Coefficient (log-odds): ≈ 0.45
- Odds ratio: exp(0.45) ≈ 1.57
- → ~57% higher odds of a high-AQI day per +1 SD wildfire exposure
- Interpretation: strongest risk signal appears with ~3-day delay

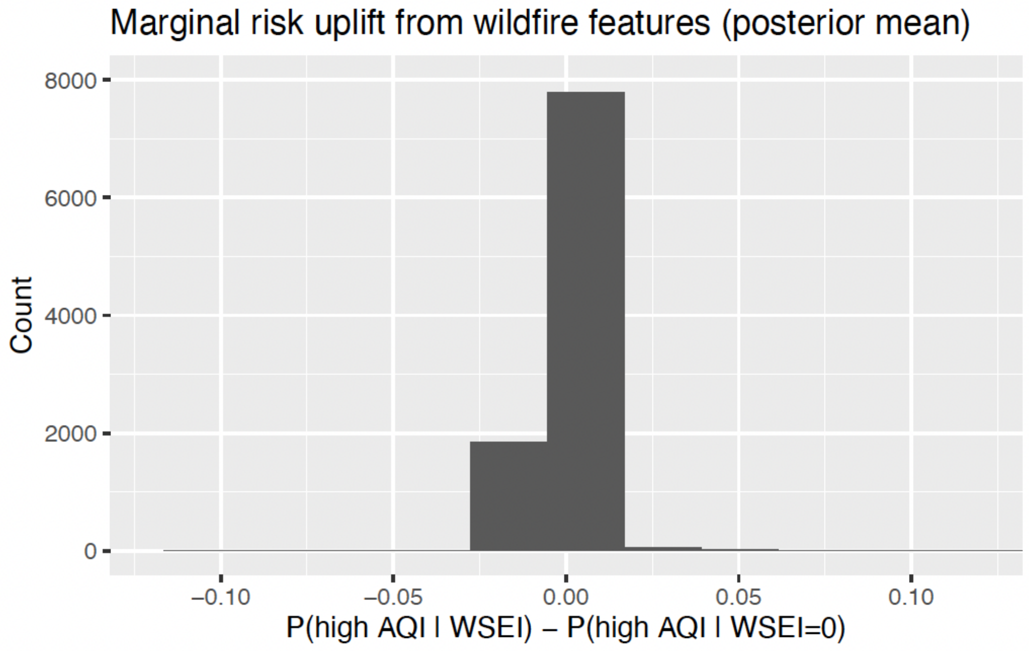
Diagnostic A: Calibration of the Risk Score

- Bin predictions into deciles of probability of high AQI next day
- Near-diagonal points → probabilities are reasonably calibrated

Diagnostic B: Wildfire-driven Risk Uplift

$\Delta = P(\text{High AQI} \mid \text{WSEI}) - P(\text{High AQI} \mid \text{WSEI} = 0)$

- Compute wildfire uplift:
- Interpretation:
  mass near 0 → wildfire has little effect on most days (good: fewer false alarms)
  positive tail → wildfire meaningfully increases risk on smoke-impacted days



Conclusion

- Wildfire exposure shows a clear positive association with next-day AQI, with the strongest and most consistent effect around a ~3-day lag; results are robust under Gaussian and Student-t errors (Student-t better reflects spike-prone AQI).
- Adding WSEI features improves performance where CS1 struggled most—high-AQI periods—by providing an additional driver that helps explain and anticipate spike days. The wildfire-aware model reduces “missing-driver” behavior relative to baselines that rely mainly on seasonality/meteorology.
- Using a top-decile next-day AQI event, the Bayesian risk model produces a probability score that is interpretable and calibratable. Wildfire features (especially lag-3) increase predicted risk, and the uplift diagnostic shows the wildfire signal activates primarily on a subset of days (good: less noise on normal days). Value to decision-making
- We can deploy a wildfire-aware early-warning indicator for Ontario: a daily risk probability + a clear explanation of why risk is elevated.